Rana Mudassir Ali

ranamudassirali9@gmail.com

Lahore, Pakistan

+923424286880

www.linkedin.com/in/rana-mudassir-ali-9a9a25238

https://github.com/ranamudassirali

Education

Bachelor of Science (BS) in Computer Science: COMSATS University Islamabad, Sahiwal Campus | 2022 – 2026
Academic Performance: CGPA: **4.0/4.0 (Patron's Double Gold Medalist)**

Intermediate: Punjab College Okara
Academic Performance: 93.8% (1032/1100)

Matriculation: Govt. Higher Secondary School Renala Khurd
Academic Performance: 92.6% (1019/1100)

Technical Skills

Programming Languages: Python, R, Kotlin, NodeJS, C++, C, SQL, Bash scripting, Assembly Language
AI/ML & Data Science: Scikit-learn for machine learning, NumPy, Data processing with Pandas, Data visualization with Matplotlib
Framework & Tools: Android Studio, Firebase, .NET development (Windows Forms), Parallel programming with OpenMP, Git repository management, NetBeans, Visual Studio, Power BI, Looker Studio, Google Colab, n8n
Web: HTML5, CSS3, JavaScript, React JS, Xampp Server

Academic Projects

Final Year Project (FYP) Corinna AI — AI-driven Sales Agent & SaaS Platform
Developed an AI-powered SaaS marketing platform with an automated sales representative for real-time customer engagement and lead generation.
Integrated an intelligent chatbot for FAQ handling, automated appointment scheduling, and secure payment processing
Tools & Technologies: Next.js, TypeScript, OpenAI API, Prisma, PostgreSQL, Clerk, Stripe, Pusher, Shadcn.

Plant Disease Classification System
Developed a CNN-based classifier to identify diseases in plant leaves using Python and TensorFlow.
Designed full preprocessing pipelines and image augmentation to enhance model training and evaluation.
Tools & Technologies: Python, TensorFlow, OpenCV, Google Colab, VS Code.

RAG-Based Chatbot System
Developed a RAG-based chatbot leveraging FAISS and Pinecone for semantic document retrieval and LLM-powered response generation.
Tools & Technologies: Python, LangChain, FAISS, Pinecone, Transformer-based Embeddings, LLM APIs

Emotion Detection System
Engineered a multi-modal system to classify human emotions from text, audio, and facial expressions.
Implemented and benchmarked Supervised, Unsupervised, and Reinforcement Learning models including CNNs and RNNs.
Tools & Technologies: Python, scikit-learn, TensorFlow, OpenCV, NLTK, librosa

Federated Learning System (Parallel & Distributed Computing)
Created a prototype for decentralized model training to demonstrate secure, privacy-preserving machine learning.
Built a client-server architecture using multi-threading and message passing in a distributed environment.
Tools & Technologies: Python, PyTorch, multiprocessing, socket programming, NumPy.

Power BI Dashboard – Social Media Insights
Constructed a dynamic dashboard to visualize key KPIs like engagement rates, impressions, and audience demographics.
Integrated diverse data sources including CSV, Excel, and APIs to create interactive visual reports.

Certifications

Data Science: 2025 Master class on Data Science Using Python, Udemy
Event Management: Computer Science Society | July 2023 – July 2024
Oracle: Oci Ai Generative Certification (Exam Applied)

Work Experience

Machine Learning & Cloud Developer | TECVEQ (Remote) ***Duration: 6 Months***
Developed and deployed Machine Learning models on GCP, utilizing cloud-based architectures for scalable data solutions.

Achievements Activities

Technical Breadth: Successfully completed 20+ micro-projects ranging from system utilities (PDF Merging, OS Modules) to interactive games.
Interests: Machine Learning, and Data Science.
Hobbies: Photography and Gaming